LECTURE-5

Macromolecules

INDRAPRASTHA INSTITUTE *of*
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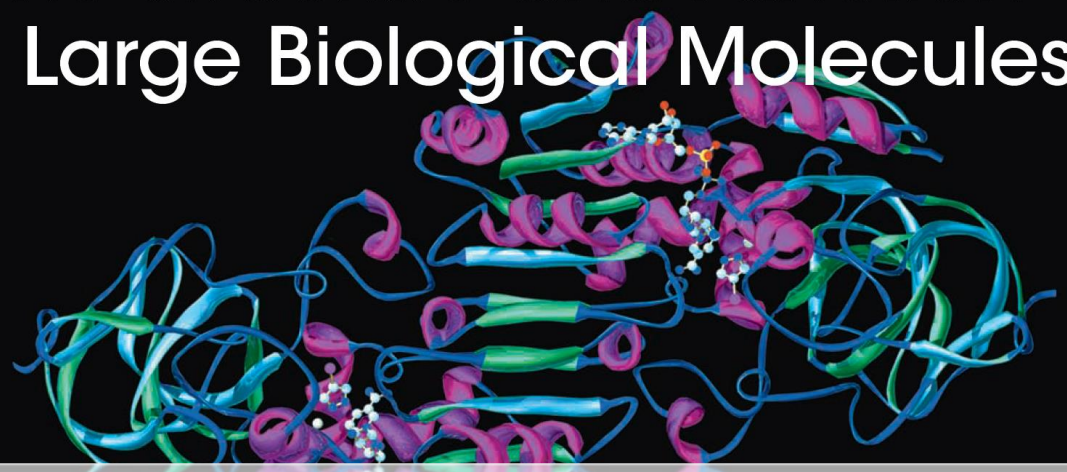
DELHI

Key concepts Macromolecules and Biopolymers

5

The Structure and Function of Large Biological Molecules



KEY CONCEPTS

5.1 Macromolecules are polymers, built from monomers

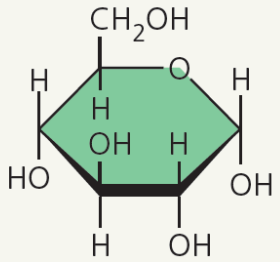
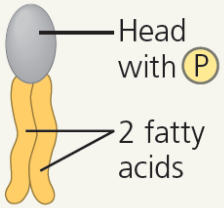
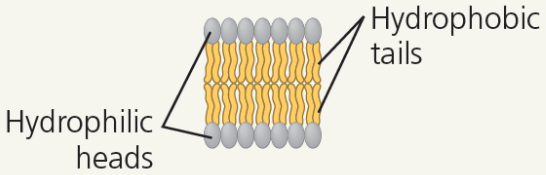
5.2 Carbohydrates serve as fuel and building material

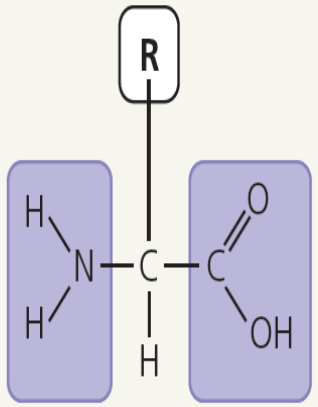
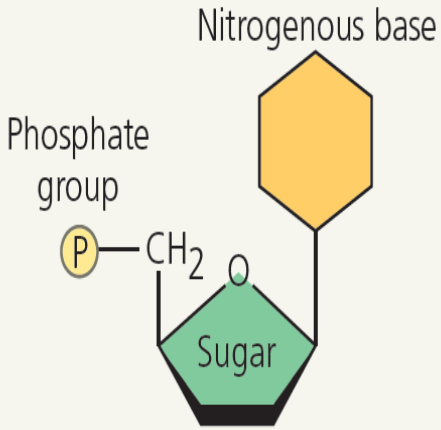
5.3 Lipids are a diverse group of hydrophobic molecules

5.4 Proteins include a diversity of structures, resulting in a wide range of functions

5.5 Nucleic acids store, transmit, and help express hereditary information

5.6 Genomics and proteomics have transformed biological inquiry and applications

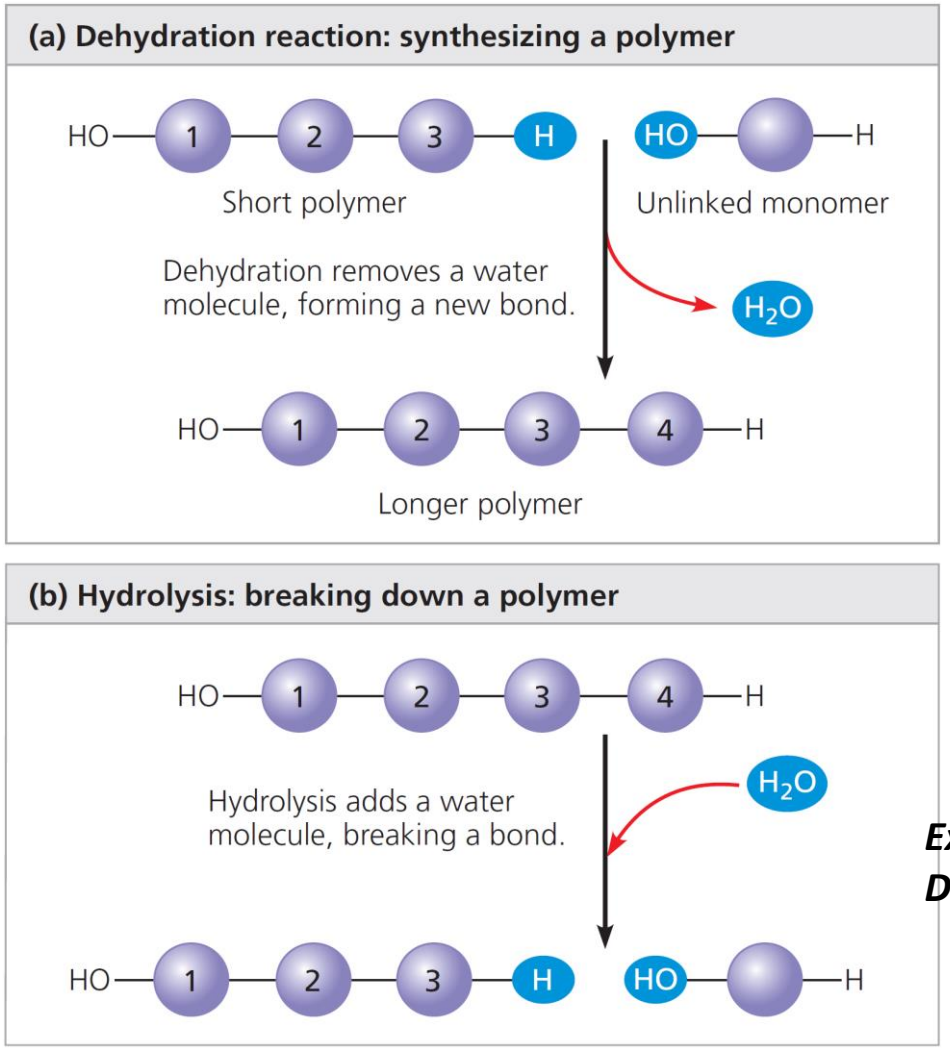
Large Biological Molecules	Components	Examples	Functions
CONCEPT 5.2 **Carbohydrates serve as fuel and building material (pp. 68–72)** ? *Compare the composition, structure, and function of starch and cellulose. What role do starch and cellulose play in the human body?*	 Monosaccharide monomer	**Monosaccharides:** glucose, fructose	Fuel; carbon sources that can be converted to other molecules or combined into polymers
		Disaccharides: lactose, sucrose	
		Polysaccharides: • Cellulose (plants) • Starch (plants) • Glycogen (animals) • Chitin (animals and fungi)	• Strengthens plant cell walls • Stores glucose for energy • Stores glucose for energy • Strengthens exoskeletons and fungal cell walls
CONCEPT 5.3 **Lipids are a diverse group of hydrophobic molecules (pp. 72–75)** ? *Why are lipids not considered to be polymers or macromolecules?*		**Triacylglycerols** (fats or oils): glycerol + 3 fatty acids	Important energy source
		**Phospholipids:** glycerol + phosphate group + 2 fatty acids	Lipid bilayers of membranes 
	 Steroid backbone	**Steroids:** four fused rings with attached chemical groups	• Component of cell membranes (cholesterol) • Signaling molecules that travel through the body (hormones)

CONCEPT 5.4 **Proteins include a diversity of structures, resulting in a wide range of functions (pp. 75–84)** ? *Explain the basis for the great diversity of proteins.*	 Amino acid monomer (20 types)	• Enzymes • Structural proteins • Hormones • Receptor proteins • Motor proteins • Defensive proteins	• Catalyze chemical reactions • Provide structural support • Coordinate organismal responses • Receive signals from outside cell • Function in cell movement • Protect against disease
CONCEPT 5.5 **Nucleic acids store, transmit, and help express hereditary information (pp. 84–87)** ? *What role does complementary base pairing play in the functions of nucleic acids?*	 Nucleotide monomer	**DNA:** • Sugar = deoxyribose • Nitrogenous bases = C, G, A, T • Usually double-stranded	Stores hereditary information
		RNA: • Sugar = ribose • Nitrogenous bases = C, G, A, U • Usually single-stranded	Various functions in gene expression, including carrying instructions from DNA to ribosomes

Macromolecules are polymers

- **Most important large molecules found in all living things**—from bacteria to elephants—can be sorted into just four main classes: **carbohydrates, lipids, proteins, and nucleic acids**.
- **Macromolecules: carbohydrates, proteins, and nucleic acids**
- A **polymer** is a long molecule consisting of many similar or identical building blocks linked by covalent bonds.
- The repeating units that serve as the building blocks of a polymer are smaller molecules called **monomers**.
- **Enzymes** are specialized macromolecules that speed up chemical reactions.

▼ **Figure 5.2** The synthesis and breakdown of polymers.



Example: Digestion

Carbohydrate serves as fuel and building material

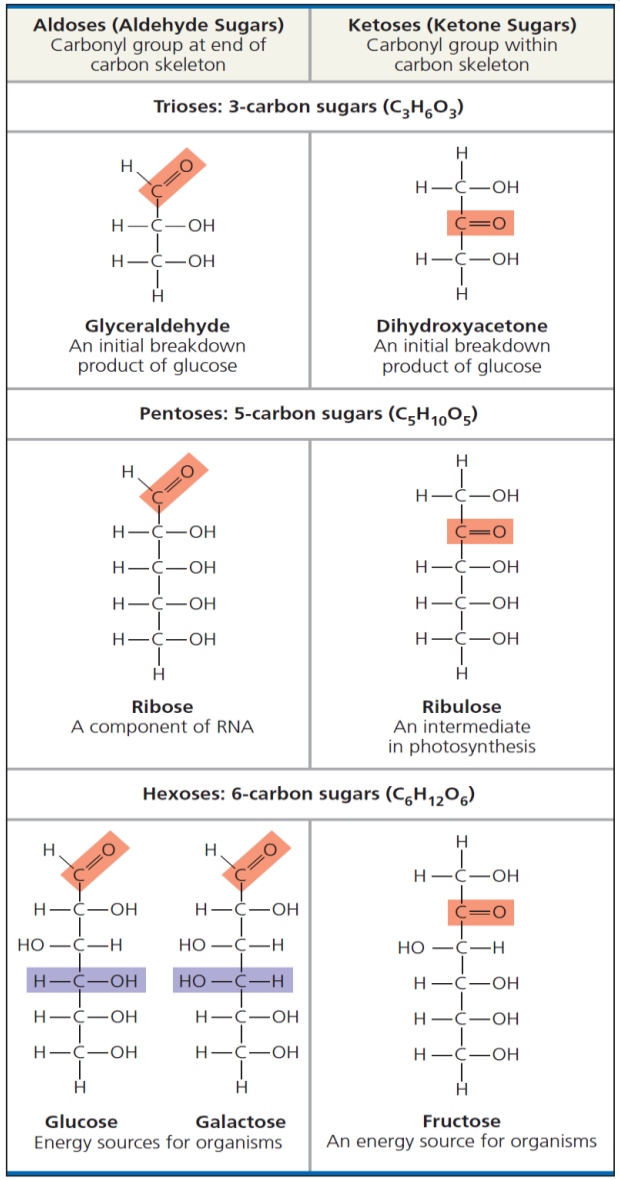
- **Carbohydrates** include sugars and their polymers.
  - ✓ The simplest carbohydrates are **monosaccharides**, or simple sugars.
  - ✓ **Disaccharides**, or double sugars, consist of two monosaccharides joined by a covalent bond.
  - ✓ **Polysaccharides** are polymers of many monosaccharides.

- **Monosaccharides** generally have molecular formulas that are some multiple of the unit $\mathbf{CH_2O}$.
  - ✓ For example, glucose has the formula $C_6H_{12}O_6$.

- Monosaccharides have a carbonyl group (>C=O) and multiple hydroxyl groups (—OH).
  - ✓ **Depending on the location of the carbonyl group, the sugar is an aldose (aldehyde sugar) or a ketose (ketone sugar)**.
  - ✓ Most names for sugars end in ***-ose***.

- Monosaccharides are also **classified by the size of the carbon skeleton**.
  - ✓ The carbon skeleton of a sugar ranges from three to seven carbons long.
  - ✓ Glucose and other six-carbon sugars are hexoses.
  - ✓ Five-carbon sugars are pentoses; three-carbon sugars are trioses.

Sugar structure and diversity



- Sugars vary in the location of their **carbonyl groups**, the length of their carbon skeletons, and the spatial arrangement around asymmetric carbons (compare, for example, the **purple portions of glucose and galactose**).

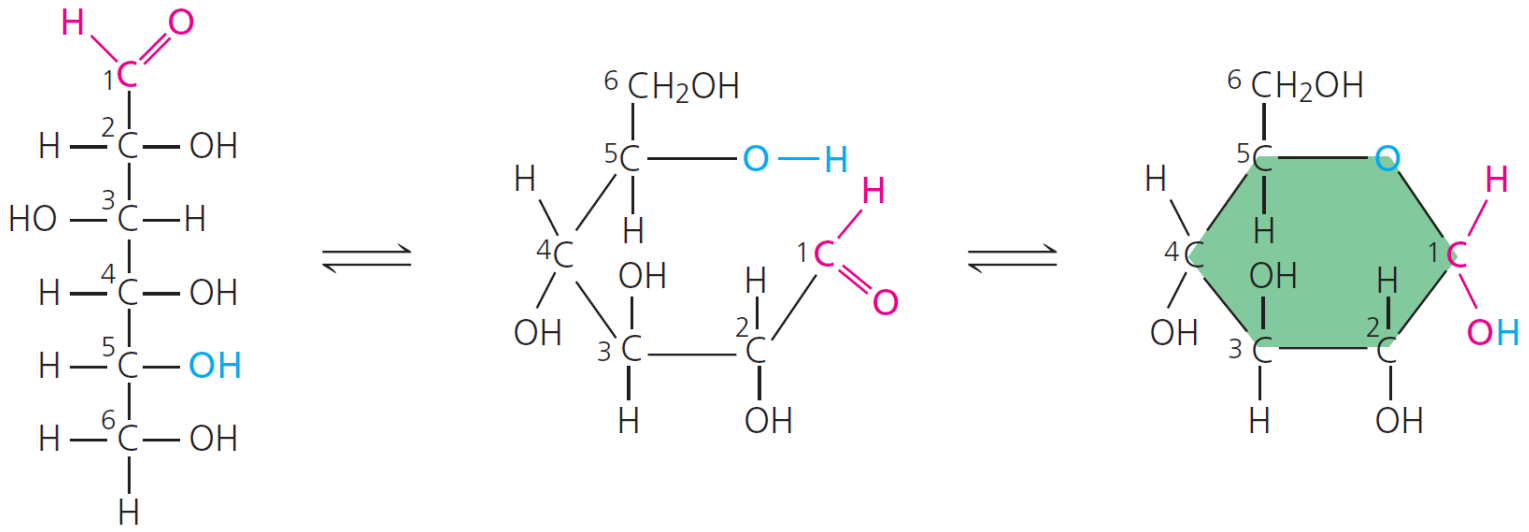
- **Another source of diversity** for simple sugars **is the spatial arrangement of their parts around asymmetric carbon atoms**.

- For example, glucose and galactose, both six-carbon aldoses, differ only in the spatial arrangement of their parts around asymmetric carbons.

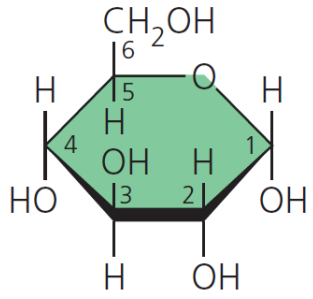
Open and closed structures

- Although glucose is often drawn with a linear carbon skeleton, **most sugars (including glucose) form rings in aqueous solution**.

- Monosaccharides, particularly glucose, are major nutrients for cellular work.
  - ✓ Cells extract energy from glucose molecules in the process of cellular respiration.



(a) Linear and ring forms. Chemical equilibrium between the linear and ring structures greatly favors the formation of rings. The carbons of the sugar are numbered 1 to 6, as shown. To form the glucose ring, carbon 1 (magenta) bonds to the oxygen (blue) attached to carbon 5.



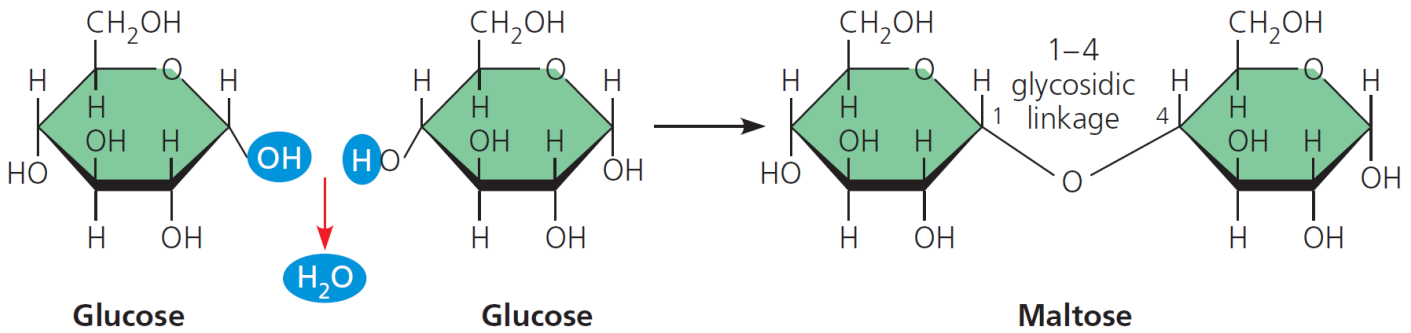
(b) Abbreviated ring structure. Each unlabeled corner represents a carbon. The ring's thicker edge indicates that you are looking at the ring edge-on; the components attached to the ring lie above or below the plane of the ring.

Disaccharides-glycosidic linkage

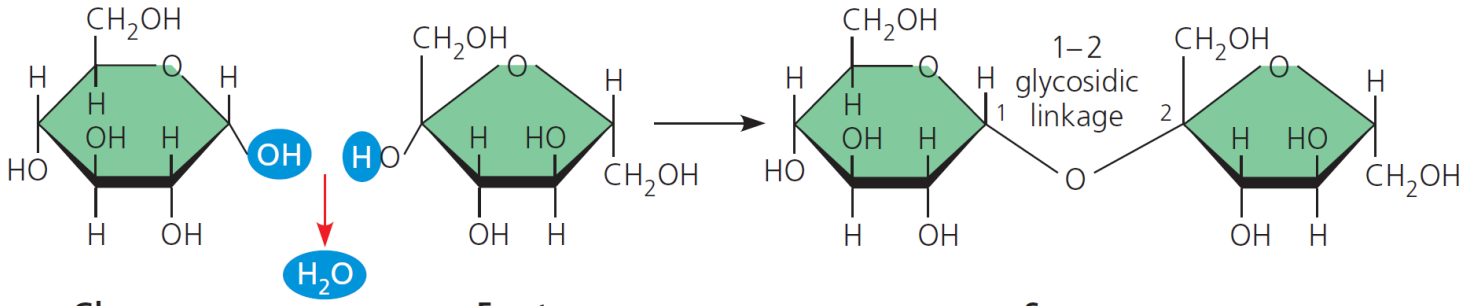
❖ Two monosaccharides can join with a **glycosidic linkage** to form a **disaccharide** via dehydration.

- ✓ Maltose, malt sugar, is formed by joining two glucose molecules.
- ✓ Sucrose, table sugar, is formed by joining glucose and fructose. Sucrose is the major transport form of sugars in plants.
- ✓

(a) Dehydration reaction in the synthesis of maltose. The bonding of two glucose units forms maltose. The 1–4 glycosidic linkage joins the number 1 carbon of one glucose to the number 4 carbon of the second glucose. Joining the glucose monomers in a different way would result in a different disaccharide.

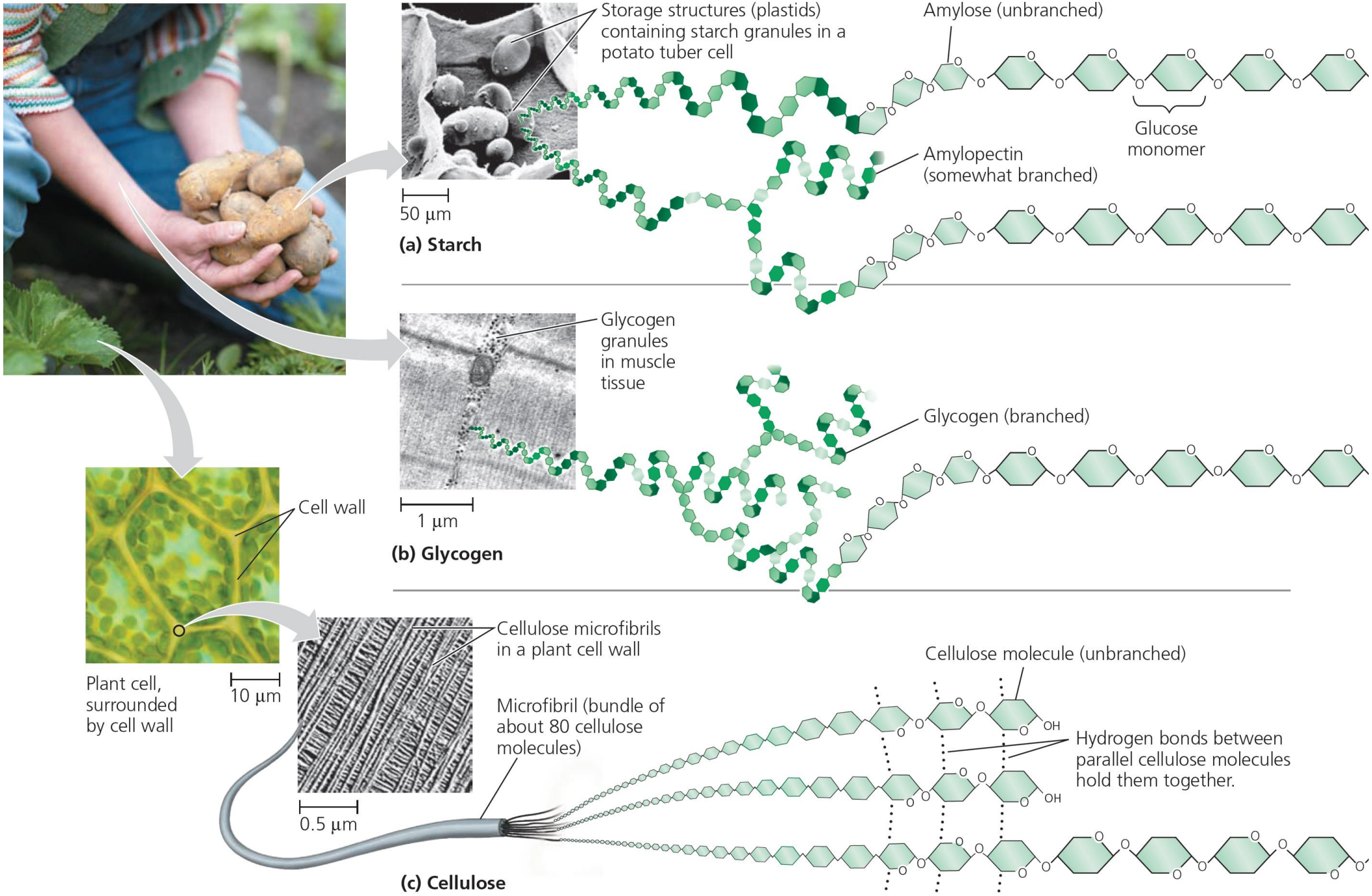


(b) Dehydration reaction in the synthesis of sucrose. Sucrose is a disaccharide formed from glucose and fructose. Notice that fructose forms a five-sided ring, though it is a hexose like glucose.



Polysaccharides

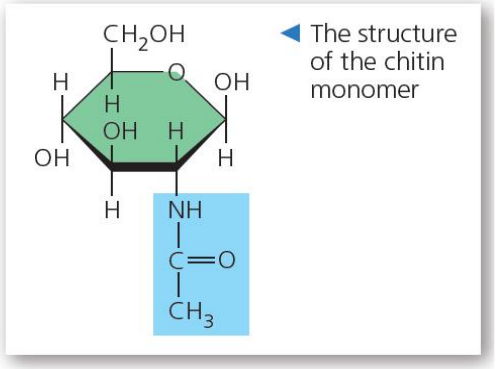
- **Polysaccharides** are polymers of hundreds to thousands of monosaccharides joined by glycosidic linkages.
  - ✓ Some polysaccharides serve for storage and are hydrolyzed as sugars are needed.
  - ✓ Other polysaccharides serve as building materials for the cell or the whole organism.
  - ✓ The architecture and function of a polysaccharide are determined by its sugar monomers and by the positions of its glycosidic linkages.
- **Starch** is a storage polysaccharide composed entirely of glucose monomers.
  - Plants store surplus glucose as starch granules within plastids.
- Animals that feed on plants, especially parts rich in starch, have **digestive enzymes** that can hydrolyze starch to glucose, making the glucose available as a nutrient for cells.
- Animals store glucose in a polysaccharide called **glycogen**.
  - ✓ Humans and other vertebrates store a day's supply of glycogen in the liver and muscles, hydrolyzing it to release glucose to meet the body's demand for sugar.


Storage structures (plastids) containing starch granules in a potato tuber cell
Amylose (unbranched)
Glucose monomer
Amylopectin (somewhat branched)
50 µm
(a) Starch
Glycogen granules in muscle tissue
Glycogen (branched)
1 µm
(b) Glycogen
Cell wall
Plant cell, surrounded by cell wall
10 µm
Cellulose microfibrils in a plant cell wall
Cellulose molecule (unbranched)
Microfibril (bundle of about 80 cellulose molecules)
Hydrogen bonds between parallel cellulose molecules hold them together.
0.5 µm
(c) Cellulose

Cellulose-important component

- **Cellulose** is a major component of the tough walls of plant cells.
  - ✓ *Plants produce almost 10^{14} kg (100 billion tons) of cellulose per year. It is the most abundant organic compound on Earth.*
- Like starch, cellulose is a polymer of glucose. However, the glycosidic linkages in these two polymers differ.
  - ✓ *The linkages are different because glucose has two slightly different ring structures.*
  - ✓ *These two ring forms differ in whether the hydroxyl group attached to the number 1 carbon is fixed above (β glucose) or below (α glucose) the plane of the ring.*
  - ✓ *Cellulose microfibrils are important constituents of **wood, paper, and cotton**.*
  - ✓ *Some microbes can digest cellulose to its glucose monomers through the use of cellulase enzymes.*
- Many eukaryotic herbivores, from cows to termites, have symbiotic relationships with cellulose-digesting prokaryotes and protists, providing the microbes and the host animal access to a rich source of energy.
  - ✓ *Some fungi can also digest cellulose.*

Chitin



◀ The structure of the chitin monomer

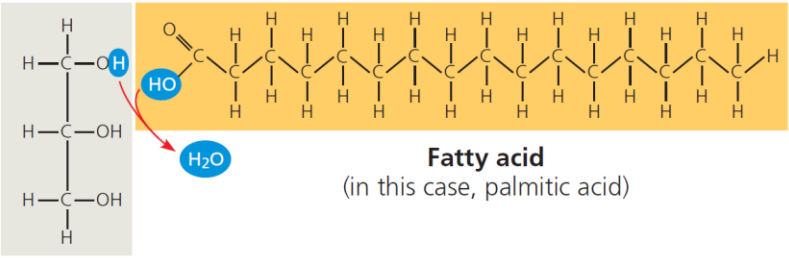
◀ Chitin, embedded in proteins, forms the exoskeleton of arthropods. This cicada is molting—shedding its old exoskeleton and emerging in adult form.

▶ Chitin is used to make a strong and flexible surgical thread that decomposes after the wound or incision heals.

▲ **Figure 5.8** **Chitin, a structural polysaccharide.**

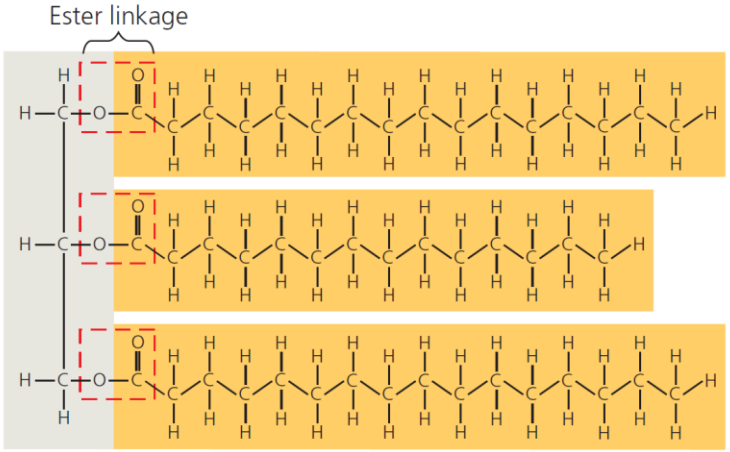
- Another important structural polysaccharide is **chitin**, found in the exoskeletons of arthropods (including insects, spiders, and crustaceans).
  - ✓ ***Chitin is similar to cellulose****, except that it has a nitrogen-containing appendage on each glucose monomer.*
  - ✓ ***Chitin*** *also provides structural support for the cell walls of many fungi.*

Lipids are diverse group of hydrophobic molecules



Glycerol

(a) One of three dehydration reactions in the synthesis of a fat



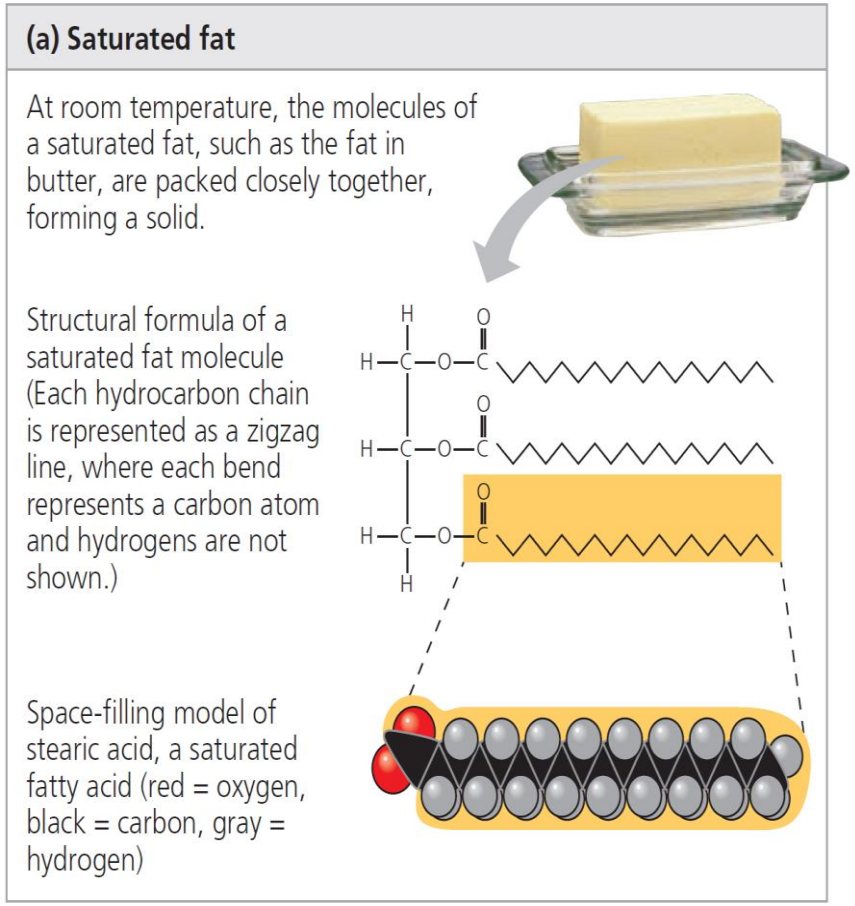
(b) Fat molecule (triacylglycerol)

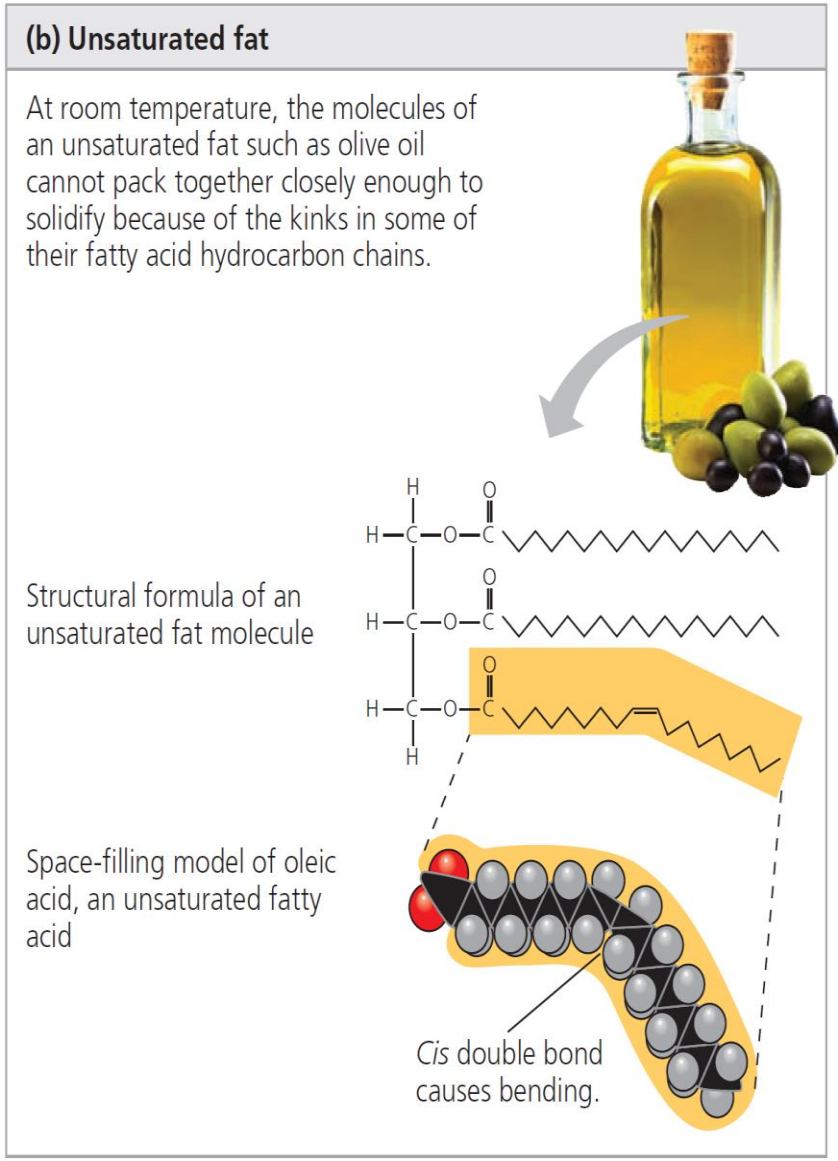
▲ **Figure 5.9 The synthesis and structure of a fat, or triacylglycerol.** The molecular building blocks of a fat are one molecule of glycerol and three molecules of fatty acids. **(a)** One water molecule is removed for each fatty acid joined to the glycerol. **(b)** A fat molecule with three fatty acid units, two of them identical. The carbons of the fatty acids are arranged zigzag to suggest the actual orientations of the four single bonds extending from each carbon (see Figure 4.3a).

- A **fat** is constructed from two kinds of smaller molecules: glycerol and fatty acids
- Unlike other macromolecules, **lipids do not form polymers**.
- The **unifying feature of lipids** is that **they have little or no affinity for water** because they consist of mostly hydrocarbons, which form nonpolar covalent bonds.
- Lipids are highly diverse in form and function.

Lipids-saturated ad unsaturated fats

Saturated: No double bonds between carbon atoms composing a chain; **Saturated fatty acid**





- *Saturated* fats and *unsaturated* fats (used in the context of nutrition) refer to the structure of the hydrocarbon chains of the fatty acids.
- **Unsaturated**: One or more double bonds. Nearly all double bonds in naturally occurring fatty acids are *cis* double bonds (causing a kink in the hydrocarbon chain).

Hydrogenated vegetable oils

- The phrase "hydrogenated vegetable oils" on food labels means that unsaturated fats have been synthetically converted to saturated fats by the addition of hydrogen.

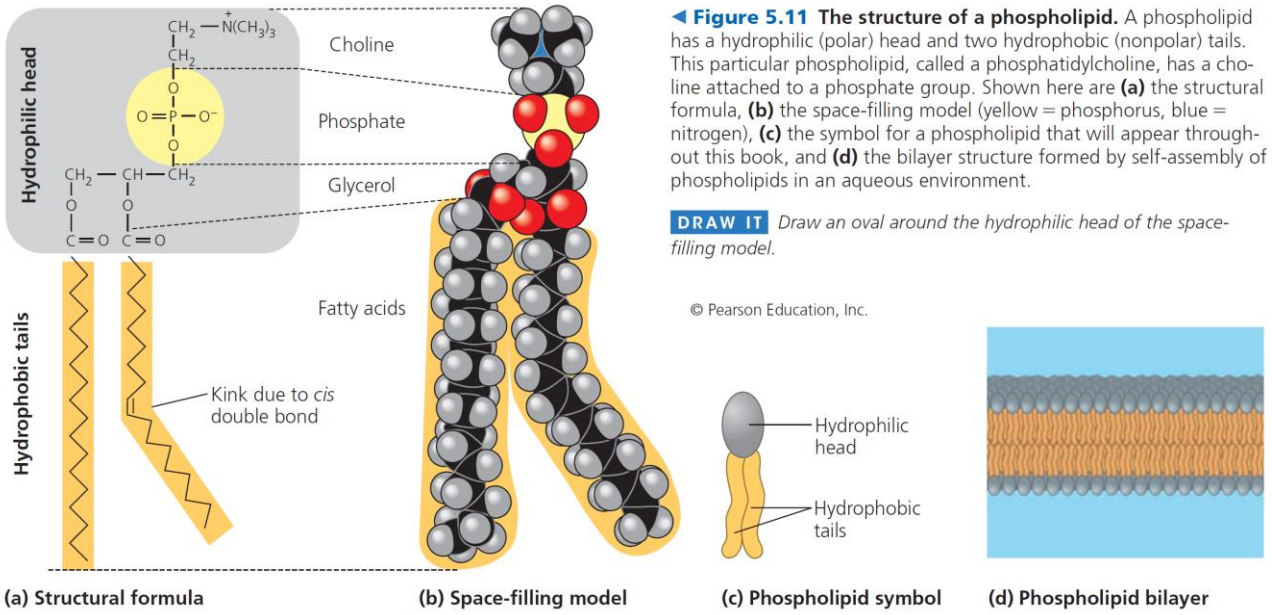
  - ✓ *Peanut butter and margarine are hydrogenated to prevent lipids from separating out as oil.*

- A **diet rich in saturated fats may contribute to cardiovascular disease** (atherosclerosis) through plaque deposits.

  - ✓ *The process of hydrogenating vegetable oils produces saturated fats and also unsaturated fats with trans double bonds. These trans fat molecules contribute more than saturated fats to atherosclerosis.*

Lipids–membranes, steroids and cholesterol

- **Phospholipids** have two fatty acids attached to glycerol and a phosphate group at the third position.
  - ✓ *The phosphate group carries a negative charge.*
  - ✓ *Additional smaller groups (usually charged or polar) may be attached to the phosphate group to form a variety of phospholipids.*
- The interaction of phospholipids with water is complex.
- **Steroids** are lipids with a carbon skeleton consisting of four fused rings.
- Different steroids are created by varying the functional groups attached to the rings.
- **Cholesterol**, an important steroid, is a component in animal cell membranes.
- Cholesterol is the precursor from which all other steroids are synthesized.
  - ✓ *Many of these other steroids are hormones, including the vertebrate sex hormones.*



◀ **Figure 5.11 The structure of a phospholipid.** A phospholipid has a hydrophilic (polar) head and two hydrophobic (nonpolar) tails. This particular phospholipid, called a phosphatidylcholine, has a choline attached to a phosphate group. Shown here are **(a)** the structural formula, **(b)** the space-filling model (yellow = phosphorus, blue = nitrogen), **(c)** the symbol for a phospholipid that will appear throughout this book, and **(d)** the bilayer structure formed by self-assembly of phospholipids in an aqueous environment.

DRAW IT *Draw an oval around the hydrophilic head of the space-filling model.*

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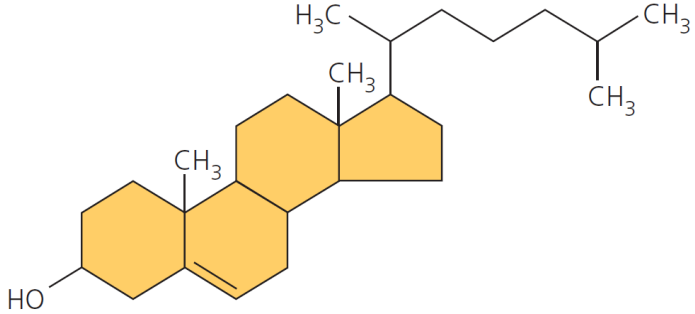

▲ **Figure 5.12 Cholesterol, a steroid.** Cholesterol is the molecule from which other steroids, including the sex hormones, are synthesized. Steroids vary in the chemical groups attached to their four interconnected rings (shown in gold).

*Although cholesterol is an essential molecule in animals, h**igh levels of cholesterol in the blood may contribute to cardiovascular disease**.*